

Aiden Tran

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Education

University of California, Berkeley

2029 - Berkeley, CA

BA in Computer Science

- **Relevant Coursework:** Structure/Interpretation of Computer Programs, Data Structures, Discrete Math/Probability Theory, Linear Algebra, Computational Astrophysics

Projects

Multiplayer Fighting Game | *Lua, Networking*

- Developed a competitive 2.5D fighting game featuring industry-standard rollback networking capable of re-simulating 12 frames of state per tick, neutralizing 150ms+ of latency for a stutter-free player experience
- Designed a scalable object-oriented architecture to manage a deterministic game loop and 60+ unique states per entity, built on Lua with a React-Lua UI layer
- **Shipped to a live player base and accumulated approximately 500k impressions**, collecting user feedback and continuously iterating with new features throughout development
- Published a technical writeup on the rollback netcode implementation (aidentran.dev/blog/rollback_netcode)

MarkovLuau | *Luau, Procedural Generation*

[GitHub](#)

- Built a Luau extension of **MarkovJunior**, giving developers a headless library of Markov chain rewrite and Wave Function Collapse algorithms to easily generate dynamic, rule-based procedural content
- Designed a programmatic DSL for writing rule sequences directly in code, alongside a parser that runs the original MarkovJunior XML models unmodified
- Built an interactive visualization tool, allowing users to preview generated outputs step-by-step and debug rule sequences
- **Authored complete documentation covering installation, getting started, worked examples, and a full API reference** (docs.aidentran.dev/markov-luau)

ML Library & LLM Inference Engine | *C++, AI/ML, CMake*

[GitHub](#)

- Developed a modular C++17 machine learning library from scratch, implementing classical models including SVMs, Linear/Logistic Regression, KNN, and Decision Trees with no external ML frameworks
- **Architected a complete LLM transformer inference stack for Llama 3.2-1B**, implementing GGUF weight loading, dequantization, KV caching, and multi-head attention entirely from scratch to run autoregressive generation
- Optimized computational throughput by implementing AVX2 SIMD vectorization for matrix-vector products, accelerating the mathematical core for real-time inference
- Implemented manual backpropagation and SMO-based SVM training with kernel caching, reducing algorithmic complexity from $O(N^3)$ to $O(N^2)$ for large-scale datasets

Real-Time Audio Signal Processor | *C++, ImGui*

[GitHub](#)

- Developed a C++ desktop application that captures live audio and analyzes frequencies in real time using a Cooley-Tukey FFT pipeline, including framing, windowing, and peak detection to identify and map fundamental frequencies to musical notes
- Rendered real-time visual feedback including waveform graphs, frequency spectrums, and detected note indicators using ImGui for a responsive and interactive UI
- Supported both real-time microphone input and static audio file analysis, processing imported recordings and returning the correlating musical note for each segment

Black Hole Spacetime Analyzer | *Python, NumPy, Matplotlib, Pandas*

[GitHub](#)

- Built a Python simulation modeling particle propagation through curved spacetime near black holes, implementing numerical integration of geodesic paths using the Schwarzschild metric to compute gravitational time dilation and redshift effects
- Integrated the AGN Black Hole Mass Database to run simulations against real supermassive black hole data, validating the model against observed astrophysical parameters
- Developed an interactive parameter system allowing users to adjust mass, radius, and observer frame and observe their impact on spacetime in real time
- Built a 3D visualization engine in Matplotlib rendering spacetime curvature surfaces and geodesic paths, graphically demonstrating event horizon boundaries for infalling objects

Experience

Key On Harmony

Jul. 2024 – Aug. 2024

Software Engineering Intern

- Developed a secure desktop ingestion client using Electron and React, replacing manual entry with a schema-validated bulk upload system

Technical Skills

Languages: C++, Python, Java, C#, Lua, TypeScript, SQL

Frameworks/Technologies: CMake, React, Node.js, Next.js, PostgreSQL, Unity, Godot, Roblox

Tools & Infrastructure: Git, Docker, Linux